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# BIRDS AND BUTTERFLIES ALONG AN URBAN GRADIENT: SURROGATE TAXA FOR ASSESSING BIODIVERSITY?

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*Abstract.* This study examines whether birds and butterflies may be used as surrogates for one another in assessing biodiversity at the community level. To do this, I compared the distribution and abundance of bird and butterfly species across an urban gradient by surveying six sites near Palo Alto, California, USA (all former oak woodlands) to see if these taxa have responded similarly to urbanization. The sites represent a gradient of urban land use ranging from relatively undisturbed to highly developed and include nature preserves, recreational areas, golf courses, residential neighborhoods, office parks, and business districts.

At the community level, the two taxa displayed similar patterns across the gradient: species richness and Shannon diversity peak at intermediate levels of development, and the oak-woodland species gradually drop out at more developed sites. These measures are highly correlated between the two groups. The two taxa differed in their patterns of total abundance, however. Butterfly abundance was highest at the preserve and decreased as the sites became more urbanized, while bird abundance peaked at a site of intermediate development. These results suggest that, on spatial scales from 1 to 10 km, the two taxa display similar patterns with regard to urbanization and that one group can be used to infer the response of the other in assessing biodiversity with these measures at the community level.

*Key words: birds; butterflies; California; conservation; development; gradient analysis; human disturbance; land use; species diversity; species richness; urbanization.*

## INTRODUCTION

The maintenance of biological diversity in the long term will require an enormous amount of effort, using a variety of strategies, directed at all levels of biological organization. Of the many options available to conservation biologists and planners, conservation of species in existing habitats will be one of the most important strategies and the selection of sites for protection will be one of the most pressing tasks (Soulé 1991, Pressey et al. 1993). Unfortunately, identifying areas of high conservation interest can require substantial amounts of time and detailed inventories of taxa, both of which may be lacking for researchers working in a discipline that has been described as ''crisis-oriented'' (Soulé 1985, Prendergast et al. 1993).

The practical implementation of conservation strategies may often be done under conditions of limited funding, information, or time. Consequently, conservation biologists have recently become interested in selecting an efficient, limited set of biological indicators for measuring and monitoring biological diversity (Kremen 1992, Pearson and Cassola 1992, Prendergast et al. 1993, Faith and Walker 1996). Much of this work has centered on the development of indexes of biotic integrity (IBIs) in aquatic systems (see Karr and Chu 1998 for an overview) with the premises that a full range of human disturbance should be examined to determine the biological response of a community, a properly selected suite of a few biological signals (such as taxon richness and relative abundance of predators) can provide reliable signals about this response, and tracking complex systems requires integrating multiple factors. Few researchers, however, have explored these ideas in terrestrial systems or investigated how well species diversity values in different groups are correlated at a local scale (Ehrlich 1996), a scale and level of diversity at which managers in the field must work.

Noss (1990) suggests that biodiversity should be monitored at multiple levels of organization and at multiple levels of spatial and temporal scales. His recommendations for indicator species include factors such as sensitivity to environmental change, wide geographic distribution, cost effectiveness in sampling, and relevance to ecologically significant phenomena. Two taxa that meet these criteria are butterflies and birds.

Many researchers (Gilbert 1980, 1984, Pyle 1980, Brown 1982, Murphy et al. 1990, Kremen 1992) have suggested that butterflies are well suited as indirect measures of environmental variation because they are

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sensitive to local climate and light levels (Watt et al. 1968, Ehrlich et al. 1972, Weiss et al. 1987, 1988). Butterfly diversity may also provide surrogate measures of plant diversity because they are directly dependent on plants, often in highly coevolved situations (Ehrlich and Raven 1964).

Others have suggested that birds, too, may serve as good indicators of environmental change. Birds are susceptible to landscape-level changes in the environment such as habitat fragmentation (Wilcove 1985). Some species are predators and, consequently, may be good indicators of subtle changes that occur at lower trophic levels, such as those that may occur through the bioaccumulation of organochlorine pesticides. Birds are relatively easy to monitor for population-level changes in breeding success and survival rates, which may reflect changes in the environment (Baillie 1991). Lastly, they may be good indicators of environmental change because extensive long-term data on bird distribution and abundance are available from Christmas Bird Counts and Breeding Bird Surveys (Robbins et al. 1986, Root 1988).

The purpose of this paper is to determine whether birds and butterflies may serve as surrogates for one another in assessing community-level biodiversity on a geographic scale of 1–10 km. To do this, I examined the effects of land use on two groups of organisms, the bird and butterfly communities of the oak woodland of central California, at a practical scale (from the perspective of natural resource managers). I used two groups of well-known organisms where the taxonomy and identification is straightforward. I examined them at the level of species diversity, which does not require in-depth analysis at either molecular or population levels. I conducted this study at a geographic scale at which land managers work, namely by examining patches of habitat within kilometers of one another. Finally, I analyze it within the context of an urbanizing environment, a situation that is becoming increasingly more common in the United States (Alig and Healy 1987, World Resources Institute 1994).

## Methods

### *Sites*

The six sites chosen for this study were located within a circle of 3-km radius centered at Stanford University near Palo Alto, California. They are representative of typical forms of development in an urban–suburban matrix, and included a biological preserve closed to the public, an open-space recreational area used for hiking, a golf course, a residential area of detached single-family housing, an office park, and a downtown business district.

The sites were chosen to be as ecologically similar to one another as possible prior to development. Historical assessment of ecological similarity was based on the presence of coast live oak (*Quercus agrifolia*), valley oak (*Quercus lobata*), or blue oak (*Quercus douglasii*) trees >100 yr old and on accounts of predevelopment vegetation (Cooper 1926). Since ''urbanization'' is an exceedingly complex amalgamation of factors (McDonnell and Pickett 1990), the rank order of the sites from most natural to most urban initially was determined using the Delphi technique, which involves gathering an anonymous consensus on a quantitative question from a group of experts ($n = 14$) when precise data are not immediately available (Zuboy 1981, Blair 1996, Blair and Launer 1997).

### *Bird survey method*

Densities of all perching or singing birds were estimated using variable circular plots at 16 survey points within each site, in a $4 \times 4$ matrix where each point is $\geq$100 m from its nearest neighbor (Reynolds et. al 1980). Points were surveyed, during peak breeding season, a total of eight times in June and July 1992, and four times in June 1993. The open-space site was visited only six times in 1992 because it burned on 10 July 1992. A daily survey began at dawn and consisted of 5-min visits to each point until all 16 points at a site were covered, ~2 h. This method results in an estimate of absolute density (number of birds per hectare) of all species within each site (see Blair 1996).

### *Butterfly and skipper survey method*

Butterflies (Papilionidae) and skippers (Hesperiidae) (collectively referred to as butterflies here) were sampled during four periods: 24 August–1 September 1992, 22 March–19 April 1993, 25 May–9 June 1993, and 13–18 August 1993. These are periods of peak flight phenology in nearby Contra Costa County as reported by Opler and Langston (1968). Each site was divided into eight areas of ~2 ha each for a total of 48 sampling areas across the urban gradient. These areas encompassed two of the contiguous bird sampling points. They were walked twice (before and after noon, on different days) for 15 min for each of the four sampling periods. All species seen during each interval were noted and voucher specimens of species difficult to identify were captured. This method results in a measure of relative abundance based on the frequency of observation of each species, which ranges from zero (no individuals of that species seen during any sampling period at that site) to 64 (at least one individual of that species seen during each 15-min sampling period at that site), when all sampling periods are combined (see Blair and Launer 1997).

### *Community measures*

Species richness is the number of species seen at each site and is directly comparable among sites because sampling effort and area were equal at all sites. Species diversity was calculated using the Shannon index (Shannon and Weaver 1949, Magurran 1988), which reflects both species richness and relative abun-

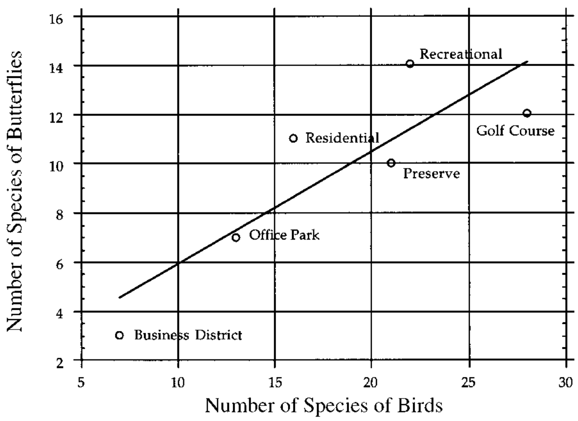


FIG. 1. Correlation between the number of species of birds and butterflies at sites of differing levels of urbanization in Palo Alto, California ($r = 0.860$, $P = 0.028$).

dance. The abundance for all birds is the sum of the densities for all species within a site. The abundance of all butterflies is the sum of the counts of all species. The species of birds and butterflies found at the biological preserve site were taken as the best representatives of the bird and butterfly communities that existed in the area prior to development—otherwise known as the minimally disturbed standard (Karr and Chu 1998). Comparison of the two taxa in species richness, diversity, abundance, and percent of woodland species was performed using Pearson correlations (SYSTAT 1992).

### *Validation of sites*

To assess whether the particular sites used in this study were typical of other sites of similar land use in the San Francisco Bay area, I conducted a validation survey of the bird life of 24 additional sites that were within 20 km of the formal study sites. These 24 sites included four different business districts, office parks, residential areas, golf courses, open-space reserves that were heavily used for recreation, and open-space reserves that were infrequently used by hikers (because there were no additional biological preserves in the area). I visited the original study sites and the 24 validation sites for 1 d each in July 1993 using the same bird sampling protocols as used in the main portion of this study. I then performed a cluster analysis using the Goodman-Kruskal gamma correlation coefficient on the count of individuals within each species (not densities) of all birds observed (SYSTAT 1992). I did not conduct a similar validation with respect to butterflies.

## RESULTS

### *Ordering of the gradient*

The ranking of the sites by the Delphi technique from most natural to most urban was: biological preserve, open-space recreational area, golf course, residential area, office park, and business district. In this order, a majority of the species of birds and butterflies displayed unimodal distributions of abundance across sites. This is reflected in the structure of the vegetation as well (Blair 1996, Blair and Launer 1997).

### *Patterns at the species level*

The patterns of changes in the distribution and abundance of species across sites were remarkably similar between the bird and butterfly communities. Of the 40 bird species in the survey, 33 had continuous, unimodal distributions (within the limits of one standard error) of abundance across the urbanization gradient. In other words, the estimated densities of these species were highest at one site along the gradient and decreased gradually at sites of either higher or lower levels of urbanization. The remaining seven species had disjunct distributions because they were not found at one or more intervening sites along the gradient (e.g., the Chestnut-backed Chickadee) (Appendix A; Blair 1996). Of the 23 butterfly species, 20 had continuous distributions and three had disjunct distributions (Appendix B; Blair and Launer 1997).

A distinct pattern of deletion and addition of species across the gradient appeared as the sites became more developed. All sites except the office park and the open-space recreational area had bird species that were unique to that site. The three least developed sites—the biological preserve, the recreational area, and the golf course—had butterfly species that were unique to those sites. Only three bird species, and no butterfly species, were found in all six sites (Figs. 1 and 2).

### *Patterns in richness, diversity, and abundance*

Trends in species richness of birds and butterflies were remarkably similar across the gradient, though the number of butterfly species was consistently lower. Both groups peaked in species richness at sites of intermediate levels of development, the birds at the golf course and the butterflies at the open space recreational

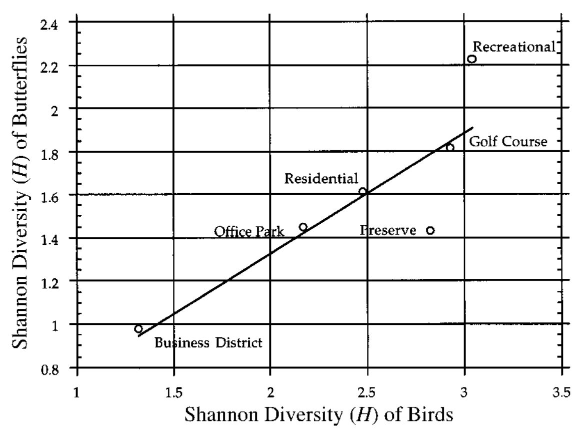


FIG. 2. Correlation between Shannon diversities of birds and butterflies at sites of differing levels of urbanization in Palo Alto, California ($r = 0.860$, $P = 0.028$).

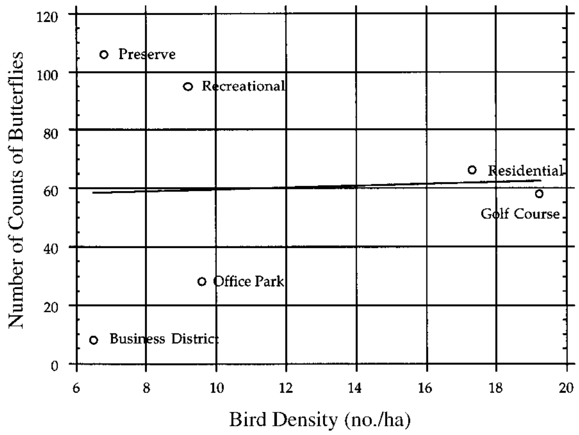


FIG. 3. Correlation between abundance of the bird and butterfly communities, measured in numbers of birds per hectare or number of observations of butterflies, at sites of differing levels of urbanization in Palo Alto, California ($r$ = 0.043, $P$ = 0.935).

area. The numbers of species of the two groups at the different sites were significantly correlated ($r$ = 0.860, $P$ = 0.028) (Fig. 1).

Species diversities of birds and butterflies were remarkably similar across the gradient as well. Both groups peaked in species diversity at the same site of intermediate development, the open space recreational area. The Shannon diversities of the two groups at the different sites were significantly correlated ($r$ = 0.860, $P$ = 0.028) (Fig. 2).

Patterns of abundance between the bird and butterfly communities were not similar. The number of butterflies counted in a site generally decreased from the preserve to the business district. The number of birds per hectare had a unimodal distribution peaking at the golf course. The abundances of the two groups at the sites were not correlated ($r$ = 0.043, $P$ = 0.935) (Fig. 3).

### *Effect on the original oak-woodland communities*

For both groups, the original oak-woodland species (defined as those found at the biological preserve) gradually dropped out along the gradient as the sites became more urbanized. Three of the bird species and none of the butterfly species found in the biological preserve were retained in the business district. The percentages of species remaining from the original assemblages decreased from the preserve to the business district and were significantly correlated between the two groups ($r$ = 0.981, $P$ = 0.001) (Fig. 4).

Curiously, both birds and butterflies showed two species that drop out early in the development gradient and reappear in the office park. The birds that reappear on the gradient—the dark-eyed junco (*Junco hyemalis*) and the cliff swallow (*Hirundo pyrrhonota*)—appear to be capitalizing on the plantings and water features in the office park. Dark-eyed juncos prefer to nest and forage under shrubs in moister woods. Cliff swallows typically breed near running water (Ehrlich et al. 1988). The butterflies—the umber skipper (*Paratrytone melane*) and the mournful dusky wing (*Erynnis tristis*)—appear to be cueing on these features as well. They are typically found near cool streamsides, openings in woodlands, and roadsides (Tilden 1965, Garth and Tilden 1986).

### *Validation of study sites*

The bird species seen at the six formal study sites (those sites upon which this study is based) are representative of the bird species found at sites of similar levels of development in the San Francisco Bay area. All of the business districts, all of the office parks, all of the golf courses, all of the preserves, and four of the open-space recreational areas were grouped by cluster analysis into distinct subsets. In addition, office parks were grouped as a distinct subset of the residential areas. One of the residential sites and one of the open-space recreational areas (not one of the formal study sites) formed a group of their own. These results suggest that, for birds, the formal study sites were more similar (in both species composition and individual species densities) to sites of similar land use than they were to sites of differing land use. This suggests that none of the formal study sites is anomalous.

## DISCUSSION

### *Butterflies and birds as indicators of diversity*

The results of this study suggest that both birds and butterflies meet many of Noss' (1990) criteria for biological indicators and that these taxa are highly similar in their response to urbanization. In particular:

1) Both taxa are distributed over a broad geographical area in that they are represented within all sites of this study, though individual species may be restricted to specific sites.

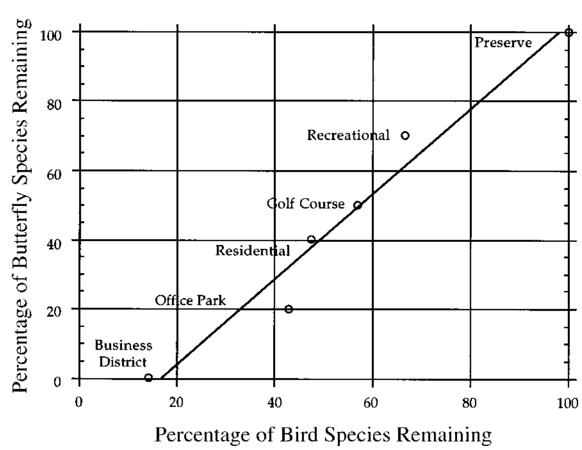


FIG. 4. The percentages of species remaining from the original assemblages of birds ($n$ = 21 species) and butterflies ($n$ = 11 species) decrease from the preserve to the business district and are highly correlated ($r$ = 0.981, $P$ = 0.001).

2) Both taxa are easy to assay because they are large, relatively simple to identify, and of high interest to the public, and both have fairly well-known life histories.

3) Both taxa appear to be sensitive enough to provide early warnings of change. Both have species that are present in the biological preserve but drop out at the next level of disturbance, the open-space recreational area.

4) Both taxa are capable of providing continuous assessment over a wide range of stress from relatively natural areas to highly urban, though the butterflies tend to be slightly less tolerant to urbanization. Few individual species are found across all levels of development.

5) Changes detectable in both taxa may be relatively independent of sample size. One-day snapshot assessments of bird life at various sites of differing land use around Palo Alto confirmed a similarity of bird species and abundance found within similar land-use types (Blair 1996).

### *Similarity of the bird and butterfly communities*

Can birds and butterflies be used as surrogates for one another in conservation planning? The similarity in response by these two communities to urbanization is remarkable. Both groups have the highest numbers of species and highest species diversity at sites of intermediate development and these measures are highly correlated between the two groups at all sites. Both of these groups have species that are found only in particular sites. Both of these groups show a gradual deletion of species from the predevelopment communities to more urbanized sites. Consequently, if conservation planners are interested in species richness, species diversity, rarity, or representativeness, either group would serve as an adequate measure for the other in these situations.

These results are particularly intriguing because they imply that taxonomic surrogates in assessing species diversity do not have to be ecological equivalents within the ecosystem. Birds and butterflies use the environment in very different ways. The diversity of birds in an area is generally correlated with the amount of structure provided by the habitat in the form of trees, shrubs, and grasses (e.g., Emlen 1974, Mills et al. 1989) and is not necessarily related to the exact plant species that are providing the structure. In contrast, butterflies respond to the environment at a very different scale, that of the individual plant and the patches in which they grow. Most butterflies and their food plants are highly coevolved and the presence or absence of individual species is correlated with the presence or absence of very specific plants (Ehrlich and Raven 1964). Consequently, one would not intuitively guess that the bird and butterfly communities, on the whole, would respond to urbanization in as similar fashion as they do.

One criterion that was not similar between the two groups was abundance. The total bird community peaked in abundance at sites of intermediate development while butterflies dropped in abundance as the sites became more urbanized. Abundance of a taxon is rarely mentioned as a criterion when evaluating a site for conservation interest (Margules and Usher 1981, Usher 1986, Spellerberg 1992) and it is falling out of favor in the creation of indexes of biotic integrity (IBIs). Originally, abundance was one criterion used in IBIs but its usefulness has been discounted in more recent efforts because population sizes vary widely naturally and these variations can easily obscure the effects of human influences on population size (Karr and Chu 1998).

Comparing these communities as biological indicators also demands a contrast of the effort required to survey these two taxa adequately. Estimating bird occurrence and abundance can be a substantial undertaking (Ralph and Scott 1981). The estimates for this study took ~300 h of early morning field work by experienced ornithologists and roughly equal amount of time in data entry and analysis. The estimates for butterfly abundance took ~100 h of midday fieldwork and a similar amount of time in data entry and analysis. The butterfly community, therefore, may be assessed more quickly and with less effort, and the assessment will achieve approximately the same results.

Butterflies are especially good indicators of changes in species diversity that occur as humans develop the landscape and excellent indicators of urbanization in particular. The butterfly community was sensitive to changes brought about by development and demonstrated this when individuals from the minimally disturbed standard community (those species found in the biological preserve) disappeared from sites of increasing development. The community, as a whole, demonstrated this with a reduction in total density across the gradient. These findings are particularly useful because these community traits may be discerned without identifying the species located at the sites but by merely relying on type identification, which can be accurate at this level (Oliver and Beattie 1996), and because the sampling may be done in a relatively short period by workers with limited training.

Some caveats should be offered with these findings. The above comparisons are predicated on the assumptions inherent in this study, namely that the potential monitoring and evaluation of sites was conducted at a scale of 1–10 km and that monitoring was to detect differences brought about by changes in land use due to urban development. These findings may not be applicable in evaluating projects at broader scales, in other ecosystem types, or under other stressors (e.g., agriculture).

For example, Prendergast et al. (1993) compared the species richness of a number of taxa in 10-km squares across the United Kingdom and found that only 12% of the species-rich hotspots for birds coincided with the species-rich hotspots for butterflies. In contrast, Pearson and Cassola (1992) compared bird and butterfly species numbers in 275-km squares across North America and found them to be significantly correlated, but this was done in an effort to explore the evolution of the two taxa in natural environments at a broad scale. Flather et al. (1997) reanalyzed Pearson and Cassola's data with the aim of controlling for the influence of latitude on species counts and found much weaker patterns of association. Flather et al. (1997) also used species-count data taken from 1-$km^2$ sampling plots in Glacier National Park, Montana, USA (data found in Debinski and Brussard 1994) and found no correlation between total butterfly counts and total bird species counts. Murphy and Wilcox (1986) found that correlation between bird and butterfly species counts varies with scale as well. In short, they found that the counts were not correlated at the scale of mountain ranges, they were at the scale of riparian canyons within a mountain range, and they were not on 1-ha plots. Taken together, the findings of these studies suggest that bird and butterfly species diversity is highly correlated within single habitat patches but not at scales of sampling that include multiple types of habitat or, perhaps, diverse levels and types of human activity (cf. Ricklefs and Schluter 1993). This study supports the idea that, on spatial scales from 1 to 10 km, birds and butterflies display similar patterns with regard to urbanization and that one group can be used to infer the response of the other in assessing biodiversity at the community level.

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## APPENDIX A

Presence of bird species in all sites in order from peak abundances in the most developed site to the most natural site (Blair 1996). This appendix is available in ESA's Electronic Data Archive: *Ecological Archives* A009-002.

## APPENDIX B

Presence of butterfly and skipper species in all sites arranged in order from peak abundance site to the most natural site (Blair and Launer 1997). This appendix is available in ESA's Electronic Data Archive: *Ecological Archives* A009-003.